AFTERSHOCK

Two robots. Evidence before action. A human at the gate.

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01 / USER AND PROBLEM

Search-and-rescue operators need to locate trapped people without sending help into an unstable route. A second collapse is the critical failure mode: a route that was clear can become unsafe before the Responder reaches the casualty.

02 / PERCEIVE, DECIDE, ACT

A light **Scout** measures the scene with simulated LiDAR, camera, thermal and acoustic channels. Map cells remain **seen, inferred or unknown**; only observed free cells support confirmed movement. Casualties carry positions, confidence and labelled simulated vitals. A heavier **Responder** follows verified routes to deliver supplies and clear only light debris.

03 / VERIFICATION AND HUMAN CONTROL

Propose > MuJoCo physics check > confidence gate > approval when risky > act. A rejected route returns to planning. Low confidence sends the Scout to re-look or refers to a human. An aftershock invalidates the old route and requires fresh evidence. Risky moves, medical guidance and emergency calls require explicit human approval. Logs connect observations, physics results and decisions.

04 / SAFETY LIMITS

No facial identification. Hazard predictions are simulated estimates, not structural engineering advice. First aid is cited guidance for a person, not diagnosis or autonomous treatment. This is a prototype of one building, one Scout and one Responder; real rescue deployment would need hardware and domain validation.

05 / ARCHITECTURE

Sim sensors > perception > planner > sim verification/actions.
One Python process; validated async topic bus. A FastAPI WebSocket sends maps, routes, reasoning and controls to the console. Sensor frames remain local. Browser writes are validated; the offline toggle extension is in review.

06 / MODEL PLACEMENT AND BUDGET

Local CPU: `local-astar` proposes core routes without a remote call. MuJoCo runs locally. The development checks used Windows 11 / Python 3.12.

Cloud features named in the spec:
`gemini-4-argon` - scene/review
`gemini-3.8-flash` - handover
`gemini-3.8-live` - commands
`gemini-3.8-flash-tts` - speech
`gemini-3.5-live-translate-preview` - casualty translation

Calls go through `coord/gem.py`. This snapshot claims no complete live-model run.

Offline goal reader: `gemma4:e2b` (Q4_K_M), Ollama 0.40.15. T15 reports an 8 GB CPU / WSL2 target, about 4.1 GB model RAM, 1.8-2.8 s warm goal reads and 19 s cold load. Budget limits: 5 GB / 4 s per step. Source: PR #55 budget.json; its demo uses a scripted physics verdict. These are lane-reported measurements; **final full-system latency is pending.**

07 / BEFORE FINAL SUBMISSION

Real-lane evaluation and one continuous demo take with an unscripted obstacle remain pending. Confirm the Sunday deadline/timezone, capture the final budget, rehearse the pitch twice and open the pack on a second machine before choosing the demo tag.

MEASURED DEVELOPMENT EVIDENCE

42 hub tests + 5 sim tests; live HTTP 200 and typed WebSocket round trip. Hub PR #47 CI passed; these checks do not establish end-to-end rescue safety.

Grid-world safe success: naive 25/30 (83.3%); T12 router 30/30 (100%). Seeds 0-29, paired layouts; reached with no collapsed cell entered or occupied at collapse.

Stand-ins: oracle map, grid dynamics, scripted collapse, test operator, no MuJoCo verification. Full-system comparison remains pending (#32 / PR #54).

Sources: docs/brief.md; docs/aftershock.md; coord/CONTRACT.md; PRs #47 and #54.